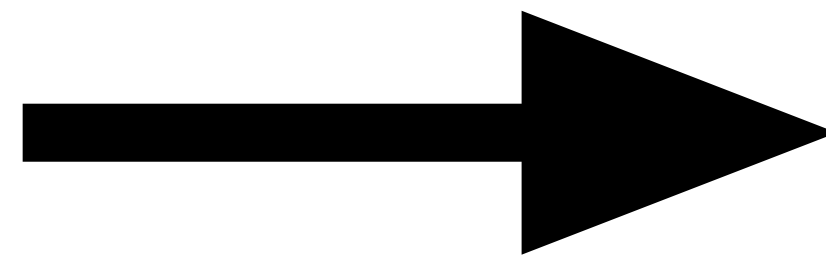


Algorithm Design: Sketches

based on the slides of 09/04/2025 and 23/04/2025

What is a Sketch ?

- U = universe of elements
- $m = |U|$
- $r : U \rightarrow [0..1]$ (real values)
- $A \subseteq U$

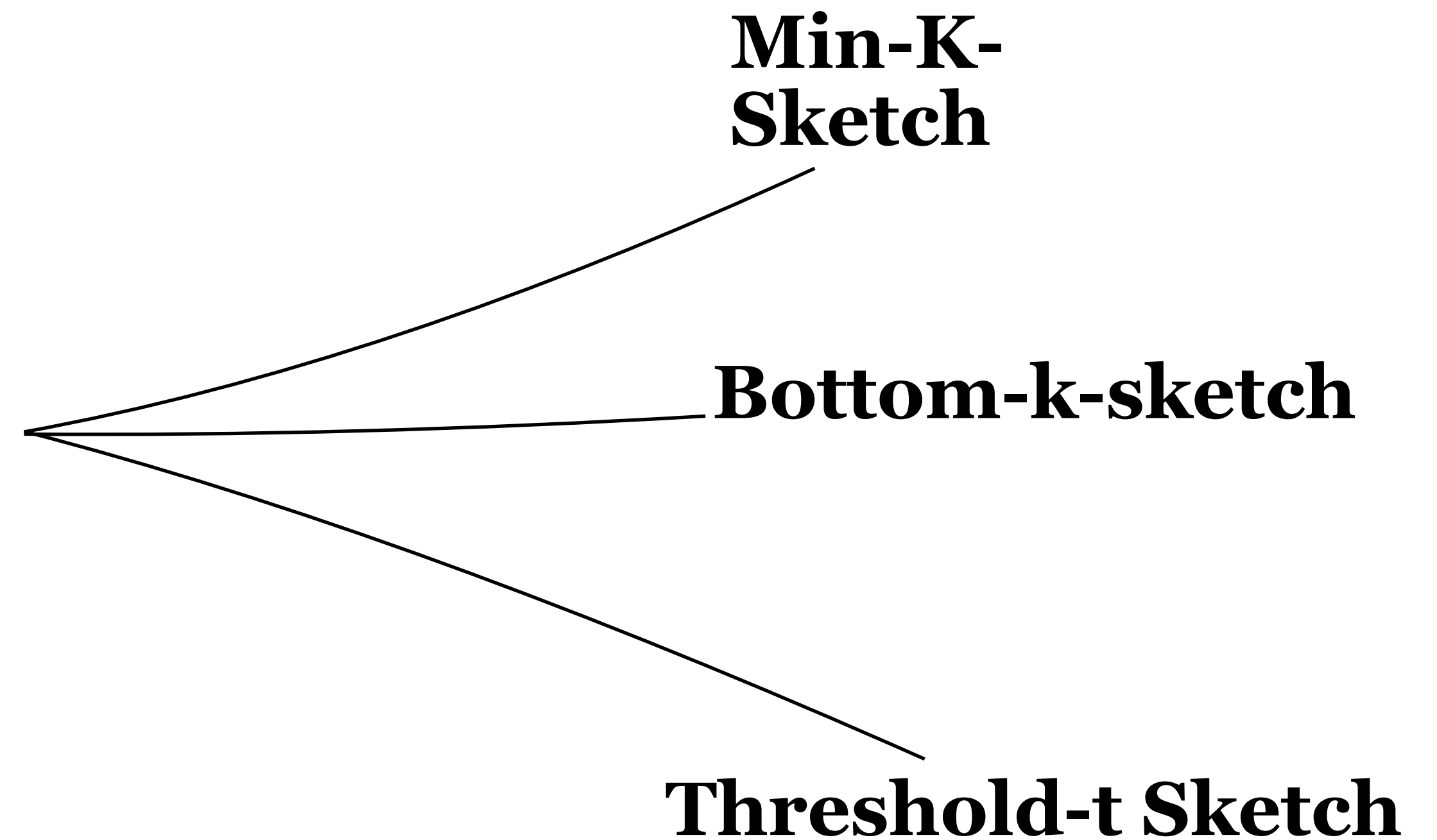


$S(A)$ is the
sketch of A

we assume that: $S(A) \subseteq A$

Operations supported by Sketches

- **INIT**: build $S(A)$ from A
- **UPDATE**: $S(A \cup \{u\})$ from $S(A)$ and u
- **UNION**: $S(A \cup B)$ from $S(A)$ and $S(B)$
- **SIZE**: estimate of $|A|$ using $S(A)$



Min-K-Sketch

K rank functions $r_1, \dots, r_k$

-INIT(A): $S(A) = \{\mathbf{argmin} \, r_i(A) : 1 \leq i \leq k\}$

-UPDATE: $S(A \cup \{u\}) = \{\mathbf{argmin}\{r_i(a_i), r_i(u)\}, 1 \leq i \leq k\}$

-UNION: $S(A \cup B) = \{\mathbf{argmin}\{r_i(a_i), r_i(b_i)\}, 1 \leq i \leq k\}$

-SIZE: $\frac{K}{\sum_{1 \leq i \leq k} r_i(a_i)} - 1$

Threshold- t Sketch

1 rank function, $t \in [0,1]$

- INIT(A):** $S(A) = \{a \in A : r(a) < t\}$
- UPDATE:** $S(A \cup \{u\}) = S(A) \cup \{u\} \iff r(u) < t$
- UNION:** $S(A \cup B) = S(A) \cup S(B)$
- SIZE :** $|S(A)| * 1/t$

Bottom-K Sketch

1 rank function

- INIT(A):** $S(A) = \{\text{smallest } K \text{ elements according to } r(A)\}$
- UPDATE:** $S(A \cup \{u\}) = \{\text{smallest } K \text{ elements of } * \text{ according to } r(*)\}$
- UNION:**
 $S(A \cup B) = \{\text{smallest } K \text{ elements of } S(A) \cup S(B) \text{ according to } r(*)\}$
- SIZE:** $(K - 1) * 1/\tau_k$ where $\tau_k = r(a)$ and a is the K -th element in $S(A)$

Example Threshold-t with Jaccardi index

$$\tilde{J}(A, B) = J(S(A), S(B))$$

$$x \in A' \cup B' : Y_r^x = 1 \iff x \in A' \cap B'$$

$$A' = S(A), B' = S(B)$$

$$\tilde{J}(A, B) = \frac{\sum_{x \in A' \cup B'} Y_r^x}{|A' \cup B'|} = \frac{|A' \cap B'|}{|A' \cup B'|}$$

$$Pr[Y_r^x = 1] = Pr[x \in A' \cap B' \mid x \in A' \cup B'] \stackrel{\text{red circle}}{=} \frac{Pr[x \in A' \cap B']}{Pr[x \in A' \cup B']} =$$

$$\frac{Pr[x \in A \cap B] * \cancel{Pr[r(x) \leq t]}}{Pr[x \in A \cup B] * \cancel{Pr[r(x) < t]}} \stackrel{\text{red circle}}{=} Pr[x \in A \cap B \mid x \in A \cup B] = J(A, B)$$

$$E[\tilde{J}(A, B)] = \frac{\sum_{x \in A' \cup B'} E[Y_r^x]}{|A' \cup B'|} = J(A, B)$$

$$\bullet Pr[E_1 \mid E_2] = Pr[E_2 \mid E_1] * \frac{Pr[E_1]}{Pr[E_2]}$$

The Milgram Experiment

Graph $G = (V, E)$ where:

- $V = \{\text{people}\}$

- $E = \{xy : x, y \in V \wedge x, y \text{ they know each other}\}$

- $d(u, v)$ = the length of the shortest path connecting u and v

- $d(u, v) - 1$ are the intermediaries between u and v

Compute Milgram's goal (average distance)

For any given $h \geq 0$,
$$N_h = \frac{|\{(u, v) \in V \times V : d(u, v) = h\}|}{n * (n - 1)}$$

$$\text{Average distance} = \sum_{u, v \in V} \frac{d(u, v)}{n(n - 1)} = \sum_{h \geq 0} h * N_h$$

Choose randomly $U \subseteq V$, we can define
$$N_h^U = \frac{|\{(u, v) \in U \times V : d(u, v) = h\}|}{|U| * (n - 1)}$$

For small U we obtain $\tilde{N}_h$ as N_h^U , running time goes from $O(|V|^2)$ to $O(|U| * |V|)$

FPTAS(ε, δ) using Azuma-Hoeffding bound ($|U| = O(\varepsilon^{-2} \log \delta^{-1})$)

$$X_i = N_h^{\{u_i\}} \implies X_i \in [0,1] \wedge a_i \leq X_i \leq b_i$$

$$U = \{u_1, u_2, \dots, u_k\}, |U| = K$$

$$N_h^U = \frac{\sum_{i=1}^K N_h^{\{u_i\}}}{K} = \frac{\sum_{i=1}^K X_i}{K} \implies E[N_h^U] = \frac{1}{K} * \sum_{i=1}^K E[N_h^{\{u_i\}}] = N_h$$

$$\mu = N_h, X_i = N_h^{\{u_i\}}, a_i = 0, b_i = 1$$

$$Pr\left[\left|\frac{\sum_{i=1}^K X_i}{K} - \mu\right| \geq \varepsilon\right] \leq 2^{-\frac{2k^2\varepsilon^2}{\sum_{i=1}^K (b_i - a_i)^2}} = 2^{-2k^*\varepsilon^2} < \delta$$

The Diameter of a Graph

Graph $G = (V, E)$

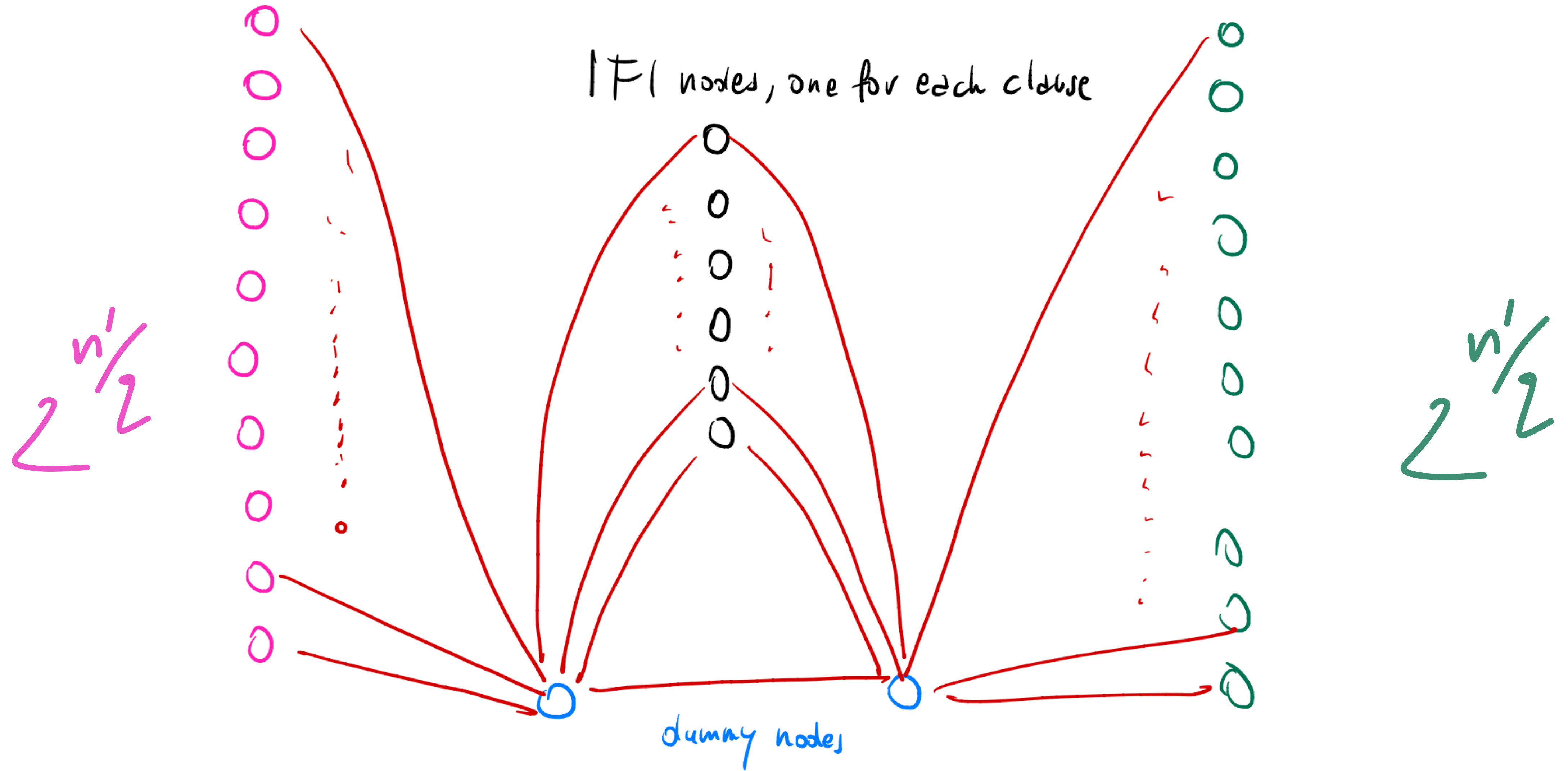
$$\text{Diameter } D = \max_{u, v \in V, u \neq v} d(x, y) = \max_{u \in V} e(u) = \max_{v \in V, v \neq u} d(u, v)$$

Baseline algo = run BFS(u)/Dijkstra(u) $\forall u \in V$, its run time is $O(nm)$ but with real-world networks, sparse graphs $m = O(n)$, the time becomes $O(n^2)$ and that is not feasible.

Strong Exponential Time Hypothesis (SETH)

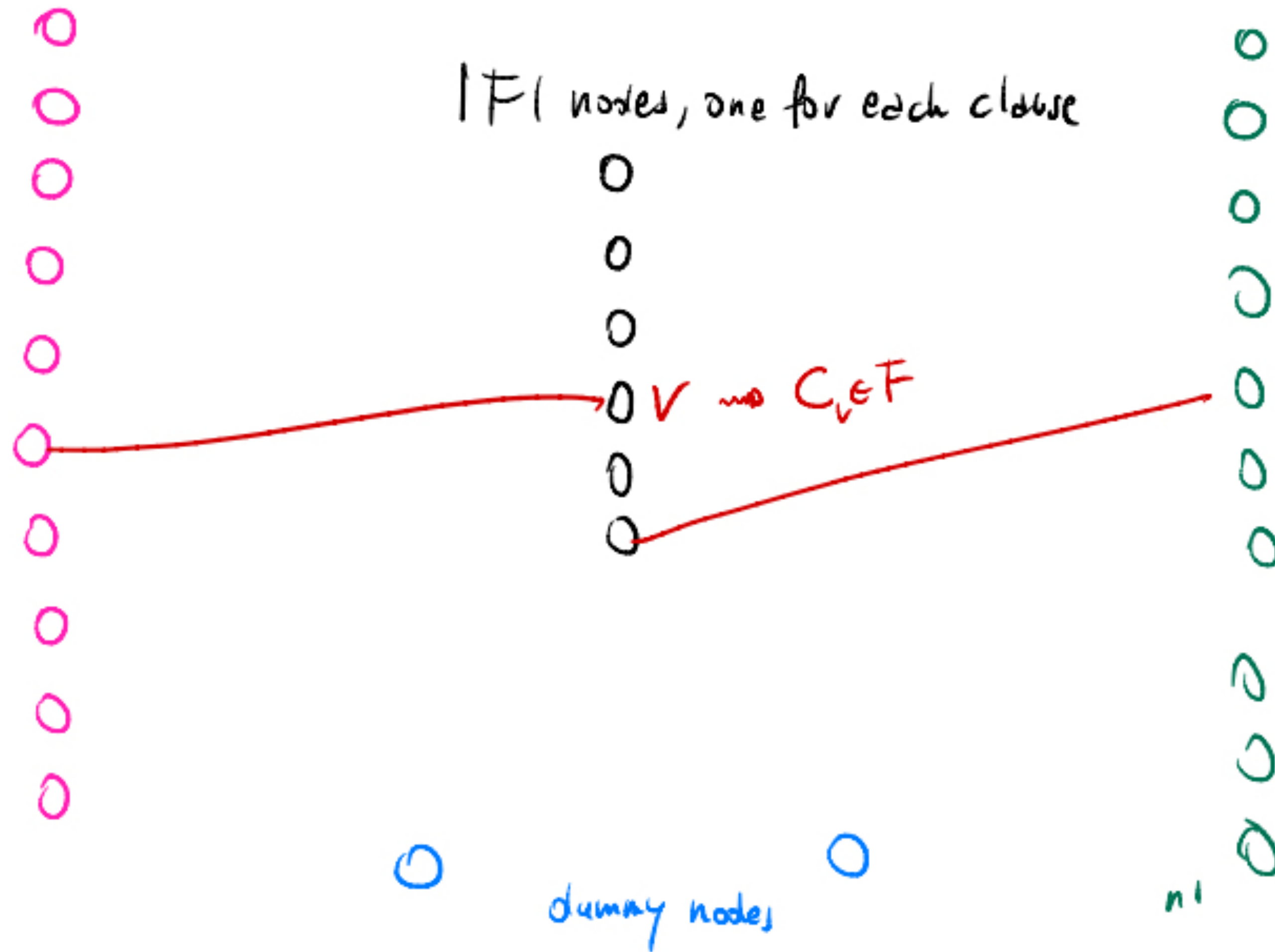
- No algorithm exists taking $O(n^{2-\varepsilon})$ time for any $\varepsilon > 0$ unless SETH is false
- literals: boolean variable
- clauses: disjunction of literals
- formula: conjunction of clauses
- assignment τ satisfies a formula $F(x_1, \dots, x_{n'}) \iff F(\tau(x_1), \dots, \tau(x_{n'})) = 1$
- SAT = Satisfiability of Boolean Formula (NP-complete)

SETH states no algorithm can solve SAT in $\tilde{O}(2^{\gamma n'})$ time for $0 < \gamma < 1$ unless $P = NP$



fixed edges $\implies D = 3$

$2^{n'/2}$



$2^{n'/2}$

fixed edges + edges depending on $F \implies |E| = O(2^{\frac{n'}{2}})$

Conclusion

F is not satisfiable $\iff \forall \tau$, there exists a clause C_v that is not satisfied
 $\iff \exists \tau_1, \tau_2$ s.t. both do not satisfy C_v $\iff \forall \tau_1, \tau_2 : d(\tau_1, \tau_2) = 2 \iff D = 2$

F is satisfiable $\iff D = 3$ by construction

Given any F, we build the previous graph with $n = O(2^{\frac{n'}{2}})$ nodes and n' variables.
Then we apply the above alg. to decide if $D=2$ or $D=3$, time is $O(n^{2-\varepsilon}) = O(2^{\frac{2-\varepsilon}{2}n'})$.
 $\gamma < 1$ as $\varepsilon > 0 \implies$ this given $O(2^{\gamma n'})$ time alg. to solve SAT $\implies$ SETH is false